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SUPERSEDING
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MILITARY SPECIFICATION

TEST SET, SIGHT CURRENT GENERATOR SERVO 11747100, AN/TSM-100

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for one type of sight Current Generator Servo Test Set, herinafter referred to as the test set, which forms a part of the Electronics System Test Set AN/TSM-100 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4931

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MIL-C-48584 - Circuit Card Assembly, Range and Range Rate Generator
MIL-C-50652/14 - Circuit Card Assembly, Range D/A Converter
MIL-C-50652/15 - Circuit Card Assembly, Range Rate D/A Converter

STANDARDS

MILITARY

MIL-STD-109 – Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

11747100 Panel Assembly, SCG Servo Test
11747105 – Schematic Diagram, Electrical, SCG Servo Test
11729016 – Wire Runs List, SCG Servo Panel

PACKAGING DATA SHEET

PS11747100 - Packaging of Test Set, Test Panel

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) , the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11747100, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the test set shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.2.1 Circuit Card Assemblies. The Circuit Card Assembly, range rate D/A converter, shall meet the requirements of MIL-C-50652\15 . Circuit Card Assembly, range D/A converter, shall meet the requirements of MIL-C-50652/14. Circuit Card Assembly, generator, shall meet the requirements of MIL-C-48584.

3.3 Environmental service conditions.

3.3.1 Vibration. The test set shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ±10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ±10%.

3.3.2 Shock. The test set shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ±10%, 11 milliseconds ±10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

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3.3.3 Altitude.

3.3.3.1 Operating. The test set shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The test set shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The test set shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage). The test set shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating] The test set shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. The circuitry shall be in accordance with electrical schematic drawing 11747105 and wire runs list 11747106. The wiring list shall take precedence over the schematic. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.5 Performance.

3.5.1 Conditions of operation. Unless otherwise specified, the requirements of 3.5 shall be met with the conditions shown in Figure 1. Printed circuit board assemblies 11747101, 10548359, and 10548330 shall be removed for tests covered by 3.5.2 to 3.5.7. Printed circuit board assemblies shall be re-inserted for tests covered by 3.5.8 to 3.5.12.

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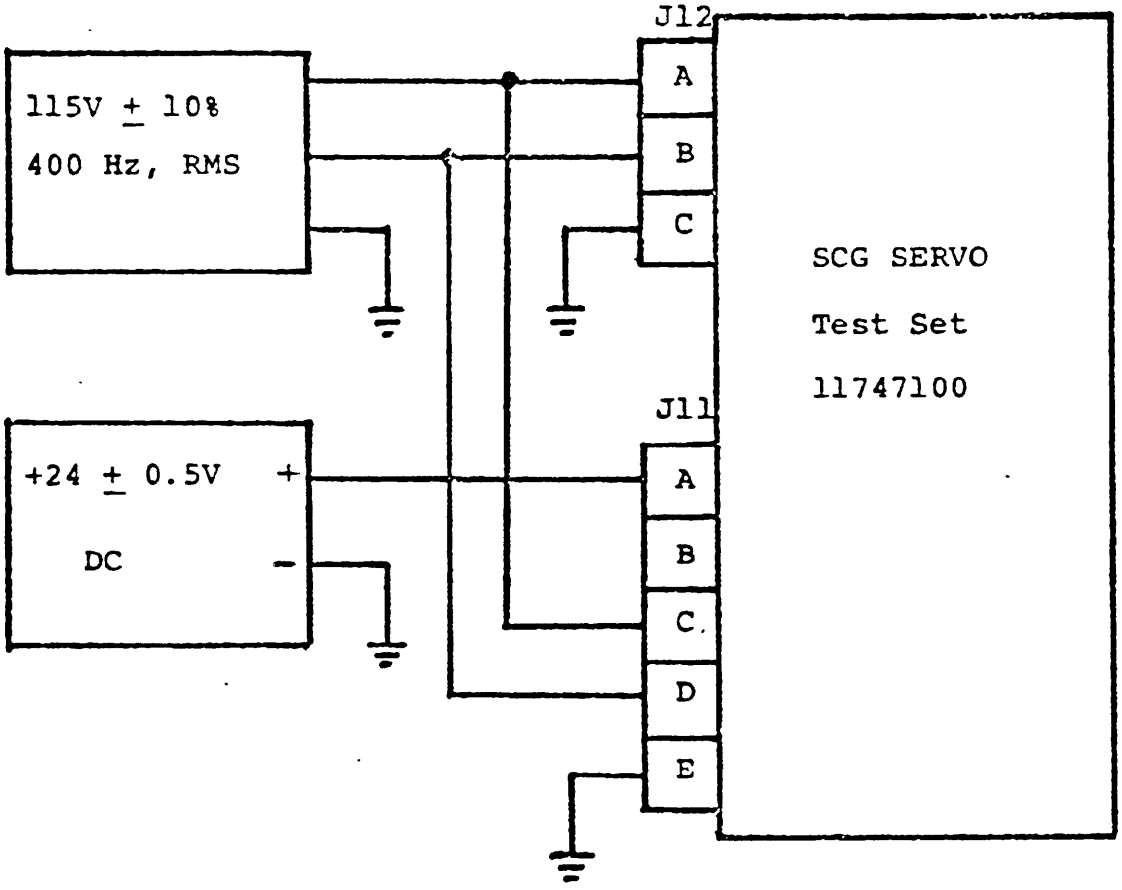


FIGURE 1. Power supply input test setup.

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3.5.2 Voltage distribution. with power switch, S25, in the On position, the following voltages shall be present at the indicated measuring points, and under the conditions specified.

a. A potential of +24.0 ±0.5V shall be present between 24 vdc test post (+), and GND 1 test post (-).

b. A potential of 115.0 ±10% Vac shall be present between ACV test post and AC-RT test post.

c. Potentials shall be present as indicated in Table I.

TABLE I. Voltage distribution

SOURCE OF VOLTAGE	MEASURING POINTS PLUS	MINUS		INDICATION
Power supply 1	TP 9-9	GND l	+5.0	+0.2 Vdc
	Jl, Pins 30,31,32	GND 1		-0.0 Vdc
	J2, Pins 30,31,32	GND 2		
	J5, Pins 30,31,32	GND 1		
Power supply 2	TP 9-6	GND 1	+15.0	+1.6 Vdc
	J5, Pins 36,37	J5, Pins 22,23,34,35		-1.2 Vdc
	J1, Pins 36,37	GND 1		
Power supply 3	TP 9-12	GND l	+15.0	+1.6 Vdc
	J1, Pins 20,21	GND 1		-1.2 Vdc
	J5, Pins 20,21	J5, Pins 22,23,34,35		
	J2, Pins 20,21	GND 2		
Power supply 4	TP 9-11	GND 1	+25.0	+1.1 Vdc
	J2, Pins 55,56	GND 2		-1.7 Vdc
Power supply 5	TP 9-10	GND 2	-25.0	+1.1 Vdc
	J2, Pins 49,50	GND 2		-1.7 Vdc

d. With the range rate switch, and the range switch in the manual position, the sequence tabulated in Table II shall occur as indicated.

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TABLE II. voltages and indications

REQUIRED OPERATION	INDICATOR LIT	MEASUREMENT VOLTAGE	TERMINALS
Sign switch on	Sign	+5 ±0.3 Vdc	TP11-7(+) & GND 1(-)
Sign switch on	Sign	+5 ±0.3 Vdc	TP11-8(+) & GND l(-)
160 M/S switch on	160 M/S	+5 ±0.3 vdc	TP11-9(+) & GND l(-)
80 M/S switch on	80 M/S	+5 ±0.3 Vdc	TP11-10(+) & GND l(-)
40 M/S switch on	40 M/S	+5 ±0.3 Vdc	TP11-11(+) & GND l(-)
20 M/S switch on	20 M/S	+5 ±0.3 Vdc	TP11-12(+) & GND 1(-)
10 M/S switch on	10 M/S	+5 ±0.3 Vdc	TP11-13(+) & GND 1(-)
5 M/S switch on	5 M/S	+5 ±0.3 Vdc	TP11-14(+) & GND 1(-)
2.5 M/S switch on	2.5 M/S	+5 ±0.3 Vdc	TP11-15(+) & GND 1(-)
5M switch on	5M	+5 ±0.3 Vdc	TP12-6(+) & GND 2(-)
10M switch on	10M	+5 ±0.3 Vdc	TP12-7(+) & GND 2(-)
20M switch on	20M	+5 ±0.3 Vdc	TP12-8(+) & GND 2(-)
40M switch on	40M	+5 ±0.3 Vdc	TP12-9(+) & GND 2(-)
80M switch on	80M	+5 ±0.3 Vdc	TP12-10(+) & GND 2(-)
160M switch on	160M	+5 ±0.3 Vdc	TP12-11(+) & GND 2(-)
320M switch on	320M	+5 ±0.3 Vdc	TP12-12(+) & GND 2(-)
640M switch on	640M	+5 ±0.3 Vdc	TP12-13(+) & GND 2(-)

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TABLE II. voltages and indications (continued)

REQUIRED OPERATION	INDICATOR LIT	MEASUREMENT VOLTAGE	MEASUREMENT TERMINALS
1280M switch on	1280M	+5 ±0.3 Vdc	TP12-14(+) & GND 2(-)
2560M switch on	2560M	+5 ±0.3 Vdc	TP12-15(+) & GND 2(-)
Target Direction Out	--	+5 ±0.3 Vdc	J5, Pin 6(+) & J5, Pin 25(-)
Target Direction In	--	+5 ±0.3 Vdc	J5, Pin 5(+) & J5, Pin 25(-)

e. With the UUT PS-1 switch and the UUT PS-2 switch in the On position, the voltages in Table III shall be present at the indicated points.

TABLE III. Voltage measurements

MEASURING POINTS		VOLTAGES
TP10-0 (+) and UUT GND 1 (-)	+15	+1.6 Vdc -1.2 Vdc
TP12-2 (+) and UUT GND 1 (-)	+15	+1.6 Vdc -1.2 Vdc
TP12-3 (+) and UUT GND 1 (-)	+15	+1.6 Vdc -1.2 Vdc
TP12-5 (+) and UUT GND 1 (-)	+15	+1.6 Vdc -1.2 Vdc
TP11-6 (+) and UUT GND 1 (-)	+15	+1.6 Vdc -1.2 Vdc
TP11-0 (+) and UUT GND 1 (-)	+15	+1.6 Vdc -1.2 Vdc
TP12-0 (+) and UUT GND 1 (-)	+15	+1.6 Vdc -1.2 Vdc

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TABLE III. Voltage measurements (continued)

MEASURING POINTS		VOLTAGES
TP11-1 (+) and UUT GND 1 (-)	+15	+1.6 Vdc -1.2 Vdc
TP9-2 and UUT VAR		+115 ±10%, RMS, 400 HZ
TP9-15 (+) and UUT GND 2 (-)	+24	+0.5 Vdc
TP9-12 (+) and UUT GND 2 (-)	-15	+1.6 Vdc -1.2 Vdc
TP9-6 (+) and UUT GND 2 (-)	+15	+1.6 Vdc -1.2 Vdc
TP9-1 (+) and TP10-6 (-)		+24 V ±10%, RMS, 400 HZ
TP9-0 (+) and TP10-6 (-)		+24 V ±10%, RMS, 400 HZ

3.5.3 Interconnections.

a. With power switch, S25, Off, the resistance measured between the test points indicated in Table IV shall be less thah 2.5 ohms. If a modifying condition is specified, it shall be performed prior to the resistance measurement.

TABLE IV. Short circuit resistance measurements

TEST POINTS	MODIFYING CONDITION
TP9-7 and TP9-5	UUT PS-2 switch on Range Ref switch in 12V Position
TP9-7 and J11-H	
TP10-1 and J1, Pins 12, 14 TP10-1 and J3, Pin 35 TP10-1 and J2, Pin 12 TP10-1 and J11-G	UUT PS-2 switch in on position
TP10-2 and J3, Pin 22 TP10-2 and J1, Pin 16 TP10-2 and J11-F	UUT PS-2 switch in on position

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TABLE IV. Short circuit resistance measurements continued)

TEST POINTS	MODIFYING CONDITION
GND 1 Terminal & J5, Pin 11	Range start switch in on position
GND 1 Terminal & TP9-14	Range & range rate switch off
GND 1 Terminal & J5, Pin 11	Range start switch off
GND 1 Terminal & TP9-14	Range & range rate switch on
GND 1 Terminal & J5, Pin 45	Range & range rate switch on
GND 1 Terminal & TP9-13	Range rate start switch off
GND 1 Terminal & J5, Pin 45	Range & range rate switch off
GND 1 Terminal & TP9-13	Range rate start switch on
J5, Pin 9 & TP11-7	Range rate switch in auto
J5, Pin 13 & TP11-8	Range rate switch in auto
J5, Pin 39 & TP11-9	Range rate switch in auto
J5, Pin 40 & TP11-10	Range rate switch in auto
J5, Pin 43 & TP11-11	Range rate switch in auto
J5, Pin 41 & TP11-12	Range rate switch in auto
J5, Pin 15 & TP11-13	Range rate switch in auto
J5, Pin 17 & TP11-14	Range rate switch in auto
J5, Pin 19 & TP11-15	Range rate switch in auto
J5, Pin 52 & TP12-6	Range switch in auto
J5, Pin 51 & TP12-7	Range switch in auto
J5, Pin 55 & TP12-8	Range switch in auto
J5, Pin 56 & TP12-9	Range switch In auto
J5, Pin 54 & TP12-10	Range switch In auto
J5, Pin 53 & TP12-11	Range switch in auto
J5, Pin 49 & TP12-12	Range switch in auto
J5, Pin 50 & TP12-13	Range switch in auto
J5, Pin 47 & TP12-14	Range switch in auto
J5, Pin 46 & TP12-15	Range switch in auto

b. With the power switch, S25, set to off position, the resistance measured between the test points indicated in Table V shall be greater than 500 K ohms. If a modifying condition is specified, it shall be performed prior to the resistance measurement.

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TABLE V. Open circuit resistance measurements.

TEST POINTS	MODIFYING CONDITION
J3, Pin 5 and J5, Pin 3	UUT PS-2 switch off
J3, Pin 5 and J5, Pin 1	
J3, Pin 35 and J1, Pin 12	UUT PS-2 switch off
J3, Pin 22 and J1, Pin 16	UUT PS-2 switch off
J3, Pin 2 and J2, Pin 41	UUT PS-1 switch off
GND 1 Terminal & J5, Pin 11	Range start switch off Range & range rate switch off
GND 1 Terminal & J5, Pin 45	Range & range rate switch off Range rate start switch off

3.5.4 Circulating fan. With the circuit shown in Figure 2 used to apply power to the fan, the fan shall operate in a smooth, quiet, reamer when the on-off switch is closed.

3.5.5 Motor-brake. With the circuit shown in Figure 3 used to provide power to the motor, the motor shall be allowed to accelerate to full speed. When switch S1 is changed to position 2, the brake shall cause a reduction in speed. When switch S1 is changed to position 3, the brake shall cause a further reduction in speed.

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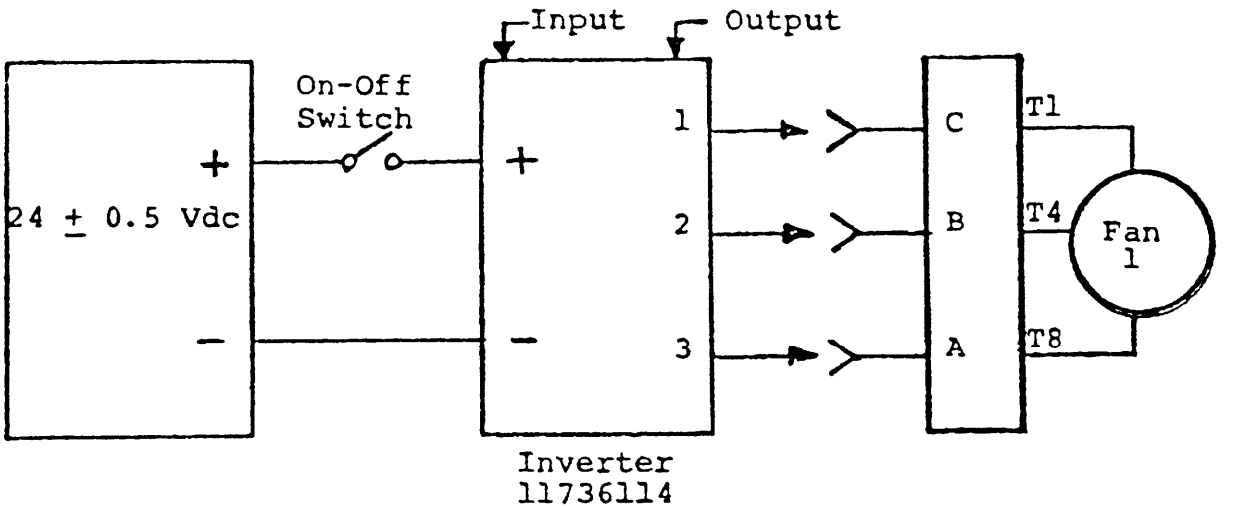


FIGURE 2. Power input to fan.

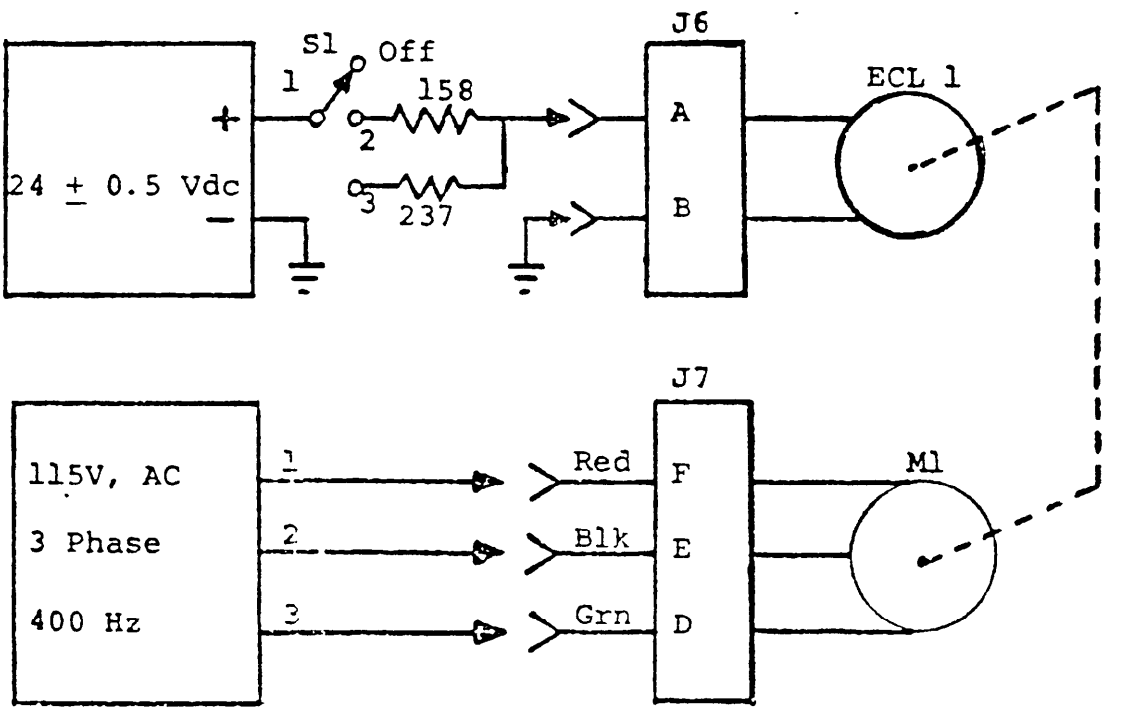


FIGURE 3. Motor brake test setup.

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3.5.6 Synchro excitation. With switches S25 and S4 alternately in the on-off positions as shown below, the outputs at test jacks shall be as indicated:

S25	S4	Outputs
Off	Off	0
Off	On	0
On	Off	0
On	On	See 3.5.7

3.5.7 Synchro signal. With the synchro locked at electrical zero degrees and S25 and S4 in the on position, the synchro output shall read 78 ± 5V when read across test jacks S1-S2 and S2-S3. The reading between R1 and S2 shall be 37 ±2 V with a short placed between R2 and S3.

3.5.8 Dynamic range output. With the power switch, S25, in the on position, the range switch, S2, and range rate switch, S1, in the auto position, and the range start switch, S28, momentarily depessed, the following signals shall be present at the indicated measuring points, and under the conditions specified.

3.5.8.1. Range output period. Range output pulses shall be present continuously for 16.0 ±0.8 sec after the range start switch is depressed and have the periods specified in Table VI. -All outputs shall be positive 50% square waves from OV to 4.0 +1.0, -1.6 Vdc. GND 2 shall be used as return for all signals. When the range pulses are present, all range indicator lamps shall be blinking in first a count-up manner and then in a count-down manner if the target direction switch, S31, is in the out position. If the target direction switch is in the in position, the lamps will blink in first a count-down manner and then in a count-up manner. All range rate indicator lamps shall at this time be extinguished.

TABLE VI. Range output period

DESCRIPTION	CHARACTERISTIC	TERMINALS
5M	16.6 ±0.4 ms	TP12-6(+) & GND 2(-)
10M	33.2 ±0.8 ms	TP12-7(+) & GND 2(-)
20M	66.4 ±1.6 ms	TP12-8(+) & GND 2(-)
40M	132.8 ±3.2 ms	TP12-9(+) & GND 2(-)
80M	265.6 ±6.4 ms	TPI2-10(+) & GND 2(-)
160M	531.2 ±12.8 ms	TP12-11(+) & GND 2(-)
320M	1.06 ±0.03 Sec	TP12-12(+) & GND 2(-)
640M	2.12 ±0.06 Sec	TP12-13(+) & GND 2(-)

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TABLE VI. Range output period (continued)

DESCRIPTION	CHARACTERISTIC	TERMINALS
1280 M	4.24 ±0.12 Sec	TP12-14(+) & GND 2(-)
2560 M	8.48 ±0.24 Sec	TP12-15(+) & GND 2(-)

3.5.9 Range analog output.

3.5.9.1 Incremental limits. With the power switch, S25, In the on position, the range switch, S2, in the manual position, and conditions specified in Table VII, voltage shall be present at TP12-1 (GND 2, RET.) as specified in Table VII.

TABLE VII. Range analog outputs

REQUIRED OPERATION (NOTE 1)	MEASUREMENT (VOLTS) NOT LESS THAN	NOT MORE THAN
5M switch on	+0.000	+0.060
10M switch on	+0.020	+0.080
20M switch on	+0.040	+0.120
40M switch on	+0.120	+0.200
80M switch on	+0.280	+0.360
160M switch on	+0.600	+0.680
320M switch on	+1.240	+1.320
640M switch on	+2.520	+2.600
1280M switch on	+5.070	+5.170
2560M switch on	+10.140	+10.340
All range switches on	+20.255	+20.665

NOTE 1: All other Range switches shall be off unless otherwise specified.

3.5.9.2 Dynamic limits. With the power switch, S25, In the on position, the range switch, S2, in the auto position, target direction switch, S31, in the out position, and the range start switch momentarily depressed, the continuous range analog voltage as shown in Figure 4, shall be present at TP11-5. With the target direction switch, S31, in the in position, and all other conditions remaining constant, the continuous range analog voltage as shown in Figure 5, shall be present at TP12-1. GND 2 shall be used as return.

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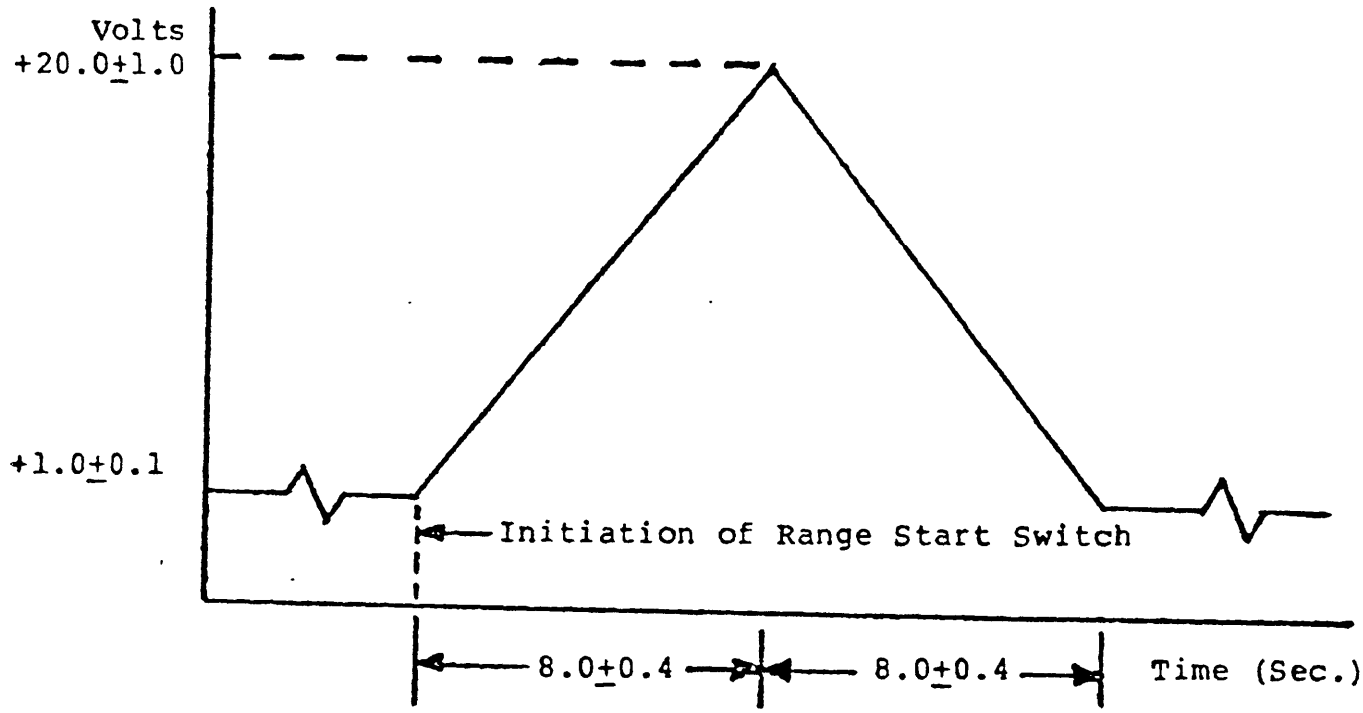


FIGURE 4. Range analog voltage.

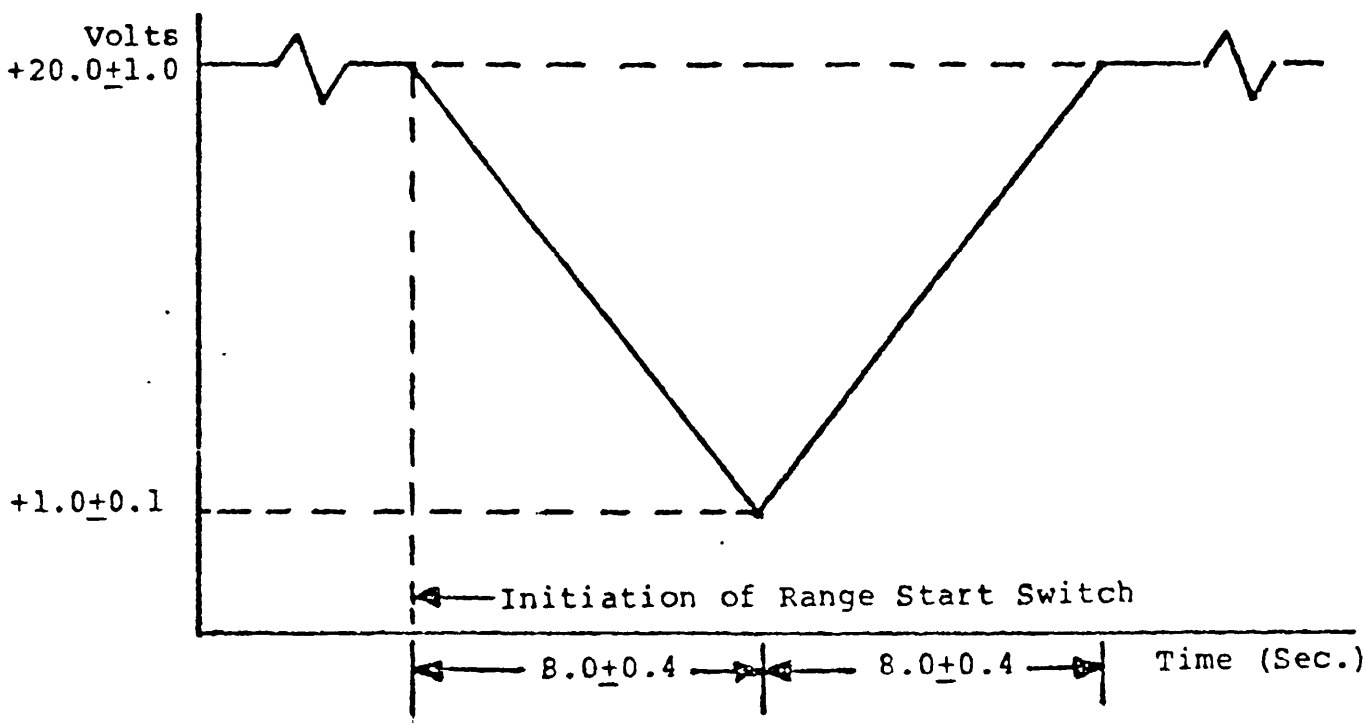


FIGURE 5. Range analog voltage.

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3.5.10 Dynamic range rate output. With the power switch, S25, in the on position, the range rate switch, S1, and range switch, S2, in the auto position and the range rate start switch momentarily depressed, the following signals shall be present at the indicated measuring points, and under the conditions specified.

3.5.10.1 Range rate output period. Range rate output pulses shall be present continuously for 16.0 ±0.8 Sec after the range rate start switch is depressed and have the periods specified in Table VIII. All outputs shall be positive 50% square waves from OV to 4.0 +1.0, -1.6, Vdc. When the range rate pulses are present, all range rate indicator lamps shall be blinking in first a count-up manner and then in a count-down manner if the target direction switch, S31, is in the out position. If the target direction switch is in the in position, the lamps will blink in first a count-down manner and then in a count-up manner. When a count-up condition exists, the Sign lamp will be illuminated and the Sign lamp will be extinguished; when a count-down condition exists, the Sign lamp will be illuminated and the Sign lamp will be extinguished. All range indicator lamps shall be at this time extinguished.

TABLE VIII. Range rate output period.

DESCRIPTION	CHARACTERISTIC	TERMINALS
2.5 M/S	132.8 ±3.2 ms	TP11-15(+) & GND l(-)
5 M/S	265.6 ±6.4 ms	TP11-14(+) & ND 1(-)
10 M/S	531.2 ±12.8 ms	TP11-13(+) & GND l(-)
20 M/S	1.06 ±0.03 Sec	TP11-12(+) & GND l(-)
40 M/S	2.12 ±0.06 Sec	TP11-11(+) & GND l(-)
80 M/S	4.24 ±0.12 Sec	TP11-10(+) & GND l(-)
160 M/S	8.48 ±0.24 Sec	TP11-9(+) & GND l(-)

3.5.11 Range rate analog output.

3.5.11.1 Incremental limits. With the power switch, S25, In the on position, the range rate switch, S1, in the manual position, and conditions specified in Table IX, voltages shall be as specified in Table IX and shall also be present at TP9-8 (GND-2, RET.) as specified in Table X.

TABLE IX. Range rate output.

DESCRIPTION	INDICATOR LAMP	MEASUREMENT VOLTAGE	TERMINALS
2.5 M/S	Off	0.0 ±0.5 Vdc	TP11-15 & GND 1
5 M/S	On	4.0 ±1.2 Vdc	TP11-14 & GND 1
10 M/S	On	4.0 ±1.2 Vdc	TP11-13 & GND 1
20 M/S	Off	0.0 ±0.5 Vdc	TP11-12 & GND 1
40 M/S	Off	0.0 ±0.5 Vdc	TP11-11 & GND 1
80 M/S	Off	0.0 ±0.5 Vdc	TP11-10 & GND 1
160 M/S	Off	0.0 ±0.5 Vdc	TP11-9 & GND 1

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TABLE X. Range rate analog outputs

REQUIRED OPERATION (NOTE 1)	MEASUREMENT (VOLTS) NOT LESS THAN	NOT MORE THAN
2.5 M/S Switch On	0.028	0.128
5 M/S Switch On	0.107	0.207
10 M/S Switch On	0.265	0.365
20 M/S Switch On	0.580	0.680
40 M/S Switch On	1.210	1.310
80 M/S Switch On	2.470	2.570
160 M/S Switch On	4.990	5.090
All Range Rate Switches On	9.900	10.100

NOTE 1: All other range rate switches shall be off unless otherwise specified.

3.5.11.2 Dynamic limits. With the power switch, S25, in the on position, the range rate switch, S1, in the auto position, target direction switch, S31, in the out position, and the range rate start switch momentarily depressed, the continuous range rate analog voltage as shown in Figure 6 shall be present at TP9-8. With the target direction switch, S31, in the in position and all other conditions remaining constant, the continuous range rate analog voltage as shown in Figure 7, shall be present at TP9-8 GND 2 shall be used as return.

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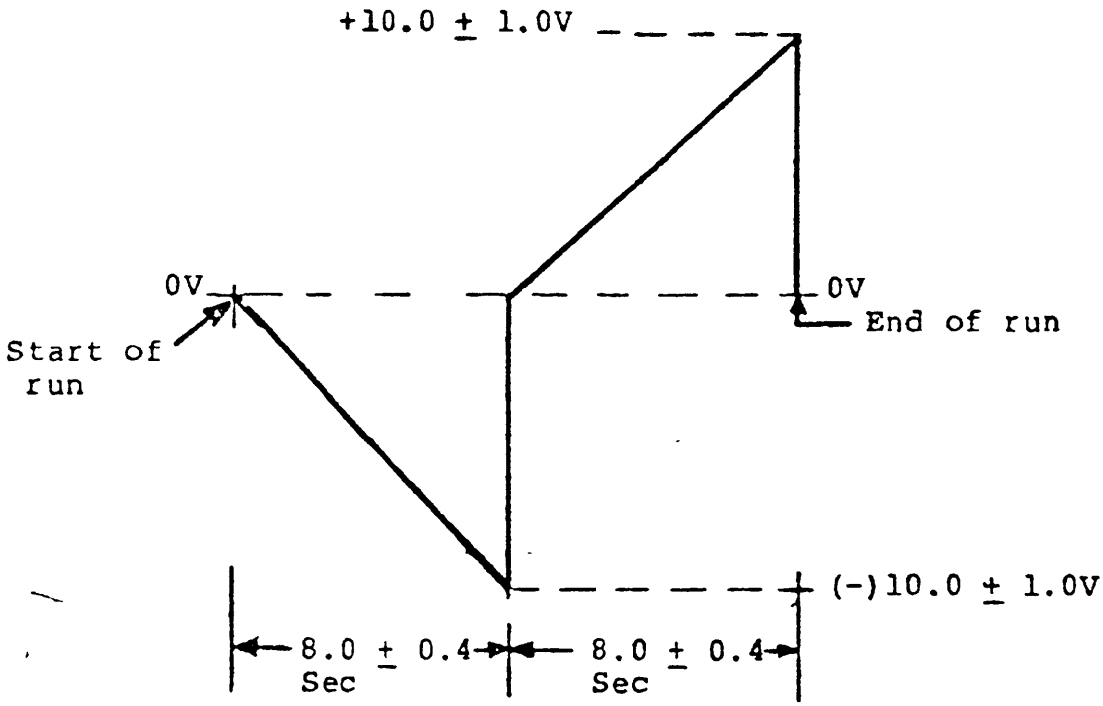


FIGURE 6. Range rate analog voltage.

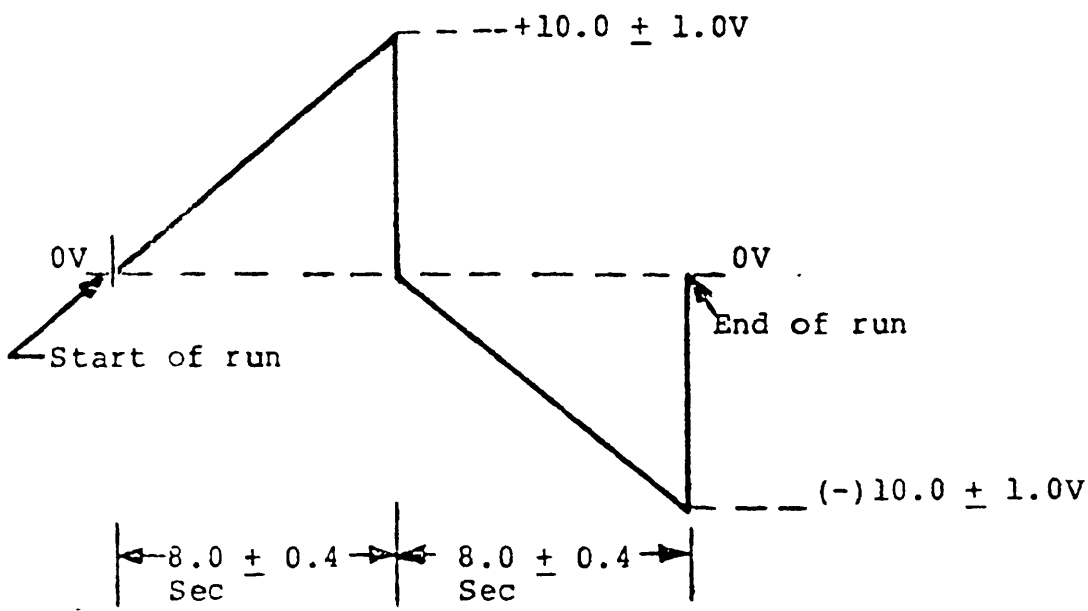


FIGURE 7. Range rate analog voltage.

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3.5.12 Range and range rate output. With the power switch, S25, in the on position, the range switch, S2, and range rate switch, S1, in the auto position, and the range and range rate switch, S30, momentarily-depressed and the target direction switch, S31, in the out position, the following conditions shall exist: The range indicator lamps shall count-up then count–down, while simultaneously the range rate indicator lamps count-up with sign lit, then count-down with sign lit.

3.5.13 Range reference outputs. With the power switch, S25, in the on position, the following voltages shall be present at the indicated points and under the conditions specified:

3.5.13.1 +20 V reference. The voltage measured at TP9-7 (+) and GND 1 (-) shall be +20.0 ±0.1 Vdc.

3.5.13.2 +12 V reference. The voltage measured at TP9-5 (+) and GND 1 (–) shall be +12.00 ±0.12 Vdc with a lK Oh, 1/2 W 1% load resistor connected across TP9 and GND 1.

3.5.13.3 Range reference switch S31. The voltage read at J3-5 (+) and GND 1 (-) shall be as in 3.5.13.1 with S3 in the +20 v position. The voltage at J3-5 and GND 1 shall be as in 3.5.13.2 with S3 set in the +12 V position.

3.5.14 RF amplifiers. With the power switch, S25, and AMPL switch, S33, in the on position, the power out and gain for AMPL 1 and AMPL 2 shall be as defined below for the inputs specified.

a. An input level of -20 dBm @ 9175 MHz and 9275 MHz input frequencies shall give an output gain of NLT 20 dB NMT 30 dB as a reference. With the output level set for +15 dB, the input level shall be not less than -15 dB and not more than -5 dB for a gain of NLT 20dB, MT 30 dB, and the gain differnce between the reference output gain and the +15 dBm output gain shall be no more than -1 dB.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Respansibility for compliance. All itiems shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+5o deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of one test set for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table XI (Functional Test) and Table XII (Special Sampling) herein and to any or all requirements of the applicable drawings.

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4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components. assemblies and subassemblies.

a. The contractor’s quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed In the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor’s proposed alternative(s) provide equal assurance, the provisions of this specification shall

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apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table XI shall be performed on a 100% basis.

TABLE XI. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
101.	Electrical	3.4	4.7.2
102.	Performance	3.5	4.7.3
Minor:		None	N/A

4.5 Special sampling. One test set out of the first so produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table XI and shall then meet the tests in Table XII. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

TABLE XII. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity	3.3.5	4.7.1.5
Minor:		None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified In the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified In "Examination" and 'Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it Is Intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. construction shall be such as to facilitate routine calibration of test equipment, Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XIII shall be used in the performance of The applicable test methods and procedures as specified in 4.7.

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1.	Power Supply	115 Vac ±10%, 400 Hz, 1 Phase, 500 milliamperes.
2.	Power Supply	0-30 Vdc, adjustable. Load regulation of 1% or better from 0 to 5 amperes. Less than 1% ripple.
3.	Digital Voltmeter	Accuracy and resolution of ±0.2% from 1 to 100 Vdc; and 1 to 200 V, 400 Hz ac. Input independence 10 megohm or greater.
4.	Ohmmeter	Accuracy and resolution of ±.05% from 1 Ohm to 10 Megohms.
5.	Test Fixture	See Figure 2. a. Inverter - 10548249 b. On-Off Switch - 5A @ 28 VdC
6.	Test Fixture	See Figure 3. a. Power Supply - 115 V ±5% RMS, 400 Hz, 3 Phase. Total harmonic waveform distortion less than 5% of the voltage of Fundamental Frequency. b. Switch S1 - Single Pole, 3 position; 10 A resistive. c. Resistor - 158 Ohms, 1%, 10 Watts d. Resistor - 237 Ohms, 1%, 5 Watts
7.	Oscilloscope	Voltage and time accuracy and resolution of 25%. Rise time: 100 nanoseconds or less. Automatic sweep speed from 1 microsecond to 100 milliseconds. Amplitude scales from 10 mV/CM to 10 V/CM.
8.	Load Resistor	1 K Ohm, l/2W, 1%.
9.	Counter	Time interval: 16 mS to 8.5 sec ±1 count. Trigger sensitivity: 0.1 V RMS min. Input impedance: 1 M Ohm.

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TABLE XIII. Test equipment list (continued)

ITEM NUMBER	DESCRIPTION
10. Signal Generator	Frequency: 9100 - 9300 MHz. Accuracy: Direct reading dial ±1%. Output : Variable -40 dBm to 0 dBm into 50 Ohms.
11. Power Meter	Frequency: 9100 - 9300 MHz. Dynamic range: -20 dBm to +10 dBm full scale.
12. Thermistor Mount	Frequency: 9100 - 9300 MHz. Power: +10 dBm direct. 200 ohm operating resistance.
13. Attenuator	Frequency: 9100 - 9300 MHz. Fixed 10dB ±0.3 dB. Rated power: 2 W average.
14. Cable	11739284-1 or equivalent.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test -performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The test set shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the test set shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock irepulses.

b. The shock pulses shall be 15 g's ±10% amplitude and 11 ms ±10% duration.

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c. The maximum g shall occur at 5.5 milliseconds ±10%.

d. The shock measuring system shall have a flat frequency response within +10% from 20 to 2,000 hertz. After the application of all the shock-impulses, the test set shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2,

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) . This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified In 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the Item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SWE) and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 Conditions of operation. Connect the power supplies (Table XIII, items 1 and 2) to Jll and J12 terminals as indicated in Figure 1 and adjust the output voltages to the values specified. Remove circuit card assemblies 10548330, 10548359, and 11747101 from the test panel. Unless otherwise specified, all tests in 4.7.3 will be performed with these conditions of operation.

4.7.3.2 Voltage distribution. Connect the digital voltmeter (Table XIII, item 3) to the test points specified in 3.5.2 and verify that the specified voltages and polarities are present at all measuring points indicated in 3.5.2, under the prescribed conditions.

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4.7.3.3 Interconnections. Connect the ohmmeter (Table XIII, item 4} to the test points specified in 3.5.3 and verify that the specified resistance is present at all measuring points indicated in 3.5.3, under the modifying conditions noted.

4.7.3.4 Circulating fan. Connect the power supply (Table XIII, item 2) and the test fixture components (Table XIII, item 5) to jack J7 of the test set per schematic of Figure 2. Operate the on-off switch and check the fan operation per the requirements of 3.5.4.

4.7.3.5 Motor brake. Construct the test fixture illustrated in Figure 3 using power supply (Table XIII, item 2) and the test fixture components (Table XIII, item 6). Energize the motor and allow the motor speed to stabilize with S1 in the off position. Check for motor speed reduction per 3.5.5 as switch S1 is changed from position 1 to alternate positions.

4.7.3.6 Synchro excitation. Turn power switch, S25, and synchro excitation switch, S4, to on position. Use voltmeter (Table XIII, item 3) to measure 115 ± 2 Vac between synchro excitation test jacks R1 and R2.

4.7.3.7 Synchro signal. Turn S25 and S4 to the positions specified in 3.5.6. Connect voltmeter to test jacks S1-S2. Reading shall be zero volt when either or both S25 and S4 are in the off position. When both switches are in the on position, the reading shall be 78 ± 5 volts. Repeat with voltmeter across S2-S3. observe same voltage.

4.7.3.7.1 Place a short between R2 and S3. Measure 37 22 volts with the voltmeter across R1 and S2 test fixture and switches S25 and S4 in off position. Disconnect instruments.

4.7.3.8 Dynamic range output. The following conditions shall be set for tests under 4.7.3.8:

a. Insert circuit card assemblies per requirements of 3.5.1. The circuit card assemblies will remain inserted for all tests covered by 4.7.3.8 to 4.7.3.13 inclusive.

b. Place the power switch, S25, in the on position.

c. Place the range rate switch, S1, in the auto position.

d. Connect the oscilloscope (Table XIII, item 7) to TP12(+) and GND 2(-).

4.7.3.8.1 Range output period. The following steps shall be followed in sequence:

a. Set TP12 to position 6 per the 5M line of Table VI.

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b. Momentarily depress the range start switch, S28, and verfiy that the signal present at the oscilloscope has the period specified on line 5M of Table VI, and meets the amplitude requirements of 3.5.8.1 as seen on oscilloscope.

c. Verify that indicator lamp conditions of 3.5.8.1 are present with target direction switch, S31, in the out position, and repeat with this switch in the in position.

d. In like manner, set TP12 to consecutive positions from TP12-7 to TP12-15 per Table VI and repeat steps b and c for lines 10M through 2560M of Table VI.

4.7.3.9 Range analog output l

4.7.3.9.1 Incremental limits. The following steps shall be followed in sequence:

a. Connect digital voltmeter (Table XIII, item 3) to TP11(+) and GND 2 (RET).

b. Place power switch, S25, in the on position.

c. Place range switch, S2, in the manual position.

d. Place TP11 in position 5.

e. Put 5M switch in on position.

f. Verify the voltage requirements of Table VII for the switch position specified. Set 5M switch to off position.

g. In like manner, activate the other switches specified in Table VII and verify the voltage requirements given.

4.7.3.9.2 Dynamic limits. The following steps shall be followed in sequence:

a. Connect oscilloscope to TP11 (+) and GND 2 (RET).

b. Set TP11 to position 5.

c. Place power switch, S25, in the on position.

d. Place range switch, S2, in the auto position.

e. Place target direction switch, S31, in out position.

f. Momentarily depress range start switch.

g. Check oscilloscope to verify the presence of a continuous range analog voltage as shown in Figure 4 and per the requirements of 3.5.9.2.

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h. Push target direction switch, S31, to in position.

i. Check oscilloscope to verify the presence of a continuous range analog voltage as shown in Figure 5 and per the requirements of 3.5.9.2.

4.7.3.10 Dynamic range rate output The following conditions shall be set for tests under 4.7.3.10:

a. Set power switch, S25, in the on position.

b. Set the range rate switch, S1, in the auto position.

c. Set the range switch, S2, in the auto position.

d. Connect the oscilloscope (Table XIII, item 7) and counter (Table XIII, Item 9) to TP11(+) and GND 1 (RET).

4.7.3.10.1 Range rate output period. The following steps shall be followed in sequence:

a. Set TP11 to position 15 per the 2.5 M/S line of Table VIII.

b. Momentarily depress the range rate start switch, and verify that the signal present at the counter has the period specified on line 2.5 M/S of Table VIII, and meets amplitude requirements of 3.5.10.1 as seen on oscilloscope.

c. Verify that indicator lamp conditions of 3.5.10.1 are present with the target direction switch, S31, in the out position, and repeat with this switch in the in position.

d. In like manner, set TP11 to consecutive positions from TP11-14 through TP11-9 per Table VIII and repeat steps b and c for lines 5 M/S through 160 M/S of Table VIII.

4.7.3.11 Range rate analog output.

4.7.3.11.1 Incremental limits. The following steps shall be followed in sequence:

a. Connect digital voltmeter to TP9(+) and GND 2 (RET).

b. Place power switch, S25, in the on position.

c. Place range rate switch, S3, in the manual position.

d. Place TP9 in position 8.

e. Set 5 M/S switch to the on position.

f. Verify the voltage requirements of Table X for the switch position specified. Put 5 M/S switch to off.

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g. In like manner, actuate the other switches in Table x and verify the voltage requirements given.

4.7.3.11.2 Dynamic limits. The following steps shall be followed in sequence:

a. connect Oscilloscope to TP9(+) and GND 2 (RET).

b. Set TP9 to position 8.

c. Place power switch, S25, in the on position.

d. Place range rate switch, S1, in the auto position.

e. Place target direction switch, S31, in the out position.

f. Momentarily depress range rate start switch.

g. Check oscilloscope to verify the presence of a continuous range rate analog voltage as shown in Figure 6 and per the requirements of 3.5.11.2.

h. push target direction switch, S31, to in position.

i. Check oscilloscope to verify the presence of a continuous range rate analog voltage as shown in Figure 7 and per the requirements of 3.5.11.2.

4.7.3.12 Range and range rate output. The following steps shall be followed in sequence:

a. Place the power switch, S25, in the on position.

b. Place range switch, S2, in the auto position.

c. Place range rate switch, S1, in the auto position.

d. Place target direction switch, S31, in the out position.

e. Momentarily depress range rate start switch, S30.

f. Verify that indicator lamps meet the requirements of 3.5.12.

4.7.3.13 Range reference outputs. With the power switch, S25, set to on, the following range reference voltages shall be present with the conditions stated in 3.5.13.

4.7.3.13.1 +20 V Reference. Connect the DVM to TP9 (+) and GND 1 (-). Set TP9 to position 7. The DVM must read +20 V reference to the limits of 3.5.13.1.

4.7.3.13.2 +12 V Reference. Set TP9 to position 5. The DVM must read +12 V to the limits and conditions of 3.5.13.2.

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4.7.3.13.3 Range reference switch S3. Connect the DVM to J3-5 (+) and GND 1 (-) on the UUT. Set range reference switch, S3 to +20 v and verfiy the +20 V output of 3.5.13.1. Set S3 to +12 V position and verify the +12 V output of 3.5.13.2.

4.7.3.14 RF Amplifiers. With the power switch, S25, and the amp switch, S33, set to on, the following tests are to be performed.

a. Set the X-Band Generator (Table XIII, item 10) dial to 9175 MHz. Connect the test cable (Table XIII, item 14) between the generator output and power meter with thermistor mount (Table XIII, items 11 and 12). Adjust the generator output for a reading of -20 dBm on the power meter. Perform 3.5.14. Repeat for 9275 MHz.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.7.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS11747100. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This test set is designed for rack-mounting into the AN/TSM-100 Electronic Systems Test Set, which is used for trouble-shooting and performance checking of specific weapon system components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Applicable stock number.
d. First Article Sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

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6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on-the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Range Rate Generator
Range D/A Converter
Synchro Signal

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A281)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-T-48593B (AR)	2 OCTOBER 1991

3 DOCUMENT TITLE
TEST SET, SIGHT CURRENT GENERATOR SERVO 11747100, AN/TSM-100

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6 SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a NAME	b TELEPHONE (*Include Area Code*)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (*Include Zip Code*) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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